

CVP ANALYSIS – QUESTIONS ONLY

Volstar Company manufactures and sells a specialized cordless telephone. The contribution format income statement for the most recent year is given.

Management is anxious to increase profit and has asked for an analysis of the following items.

Required:

Question 1

Compute the company's:

- (a) Contribution margin (CM) ratio
- (b) Variable expense ratio

Question 2

Compute the company's break-even point:

- (a) In units
 - (b) In sales dollars
- Use the equation method.

Question 3

Assume that sales increase by \$400,000 next year.

If cost behavior patterns remain unchanged, by how much will net operating income increase?

Use the CM ratio to compute your answer.

Question 4

Refer to the original data.

Assume that next year management wants the company to earn a target profit of at least \$90,000.

How many units must be sold to achieve this target profit?

Question 5

Refer to the original data.

Compute the company's margin of safety:

- (a) In dollars
- (b) In percentage form

Question 6

(a) Compute the company's degree of operating leverage (DOL) at the present level of sales.

(b) Assume that through a more intense sales effort, the company's sales increase by 8% next year. By what percentage will net operating income increase?

Use the degree of operating leverage to compute your answer.

(c) Verify your answer to part (b) by preparing a new contribution format income statement showing an 8% increase in sales.

Question 7

To increase sales and profits, management is considering using a higher-quality speaker.

- Variable costs will increase by \$3 per unit
- One quality inspector will be eliminated, reducing fixed costs by \$30,000 per year
- Sales are expected to increase by 20%

Required:

(a) Prepare a projected contribution format income statement for next year showing:

- Total
- Per unit
- Percentage of sales

(b) Compute the company's new break-even point:

- In units
- In sales dollars

(c) Would you recommend that the proposed changes be made? Explain briefly.

1 Contribution Margin Ratio & Variable Expense Ratio

Step 1: Write the formula

$$\text{Contribution Margin Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

$$\begin{aligned} &= \frac{300,000}{1,200,000} \\ &= 0.25 = \boxed{25\%} \end{aligned}$$

Variable Expense Ratio

Step 1: Formula

$$\text{Variable Expense Ratio} = \frac{\text{Variable Expenses}}{\text{Sales}}$$

$$\begin{aligned} &= \frac{900,000}{1,200,000} \\ &= 0.75 = \boxed{75\%} \end{aligned}$$

2 Break-Even Point (Equation Method)

Step 1: Write the basic CVP equation

$$\text{Sales} - \text{Variable Cost} - \text{Fixed Cost} = 0$$

(a) Break-even in Units

Step 2: Convert equation into units

$$(\text{Selling price per unit} \times Q) - (\text{Variable cost per unit} \times Q) - \text{Fixed cost} = 0$$

Step 3: Substitute values

$$(60Q) - (45Q) - 240,000 = 0$$

$$15Q = 240,000$$

$$Q = \frac{240,000}{15}$$

$$Q = \boxed{16,000 \text{ units}}$$

(b) Break-even in Sales Dollars

Step 1: Formula

$$\text{Break-even sales} = \frac{\text{Fixed Cost}}{\text{CM Ratio}}$$

$$= \frac{240,000}{0.25}$$

$$= \boxed{960,000}$$

3 Increase in NOI if Sales Increase by \$400,000

Step 1: Write the equation

$$\Delta \text{NOI} = \text{CM Ratio} \times \Delta \text{Sales}$$

Step 2: Substitute

$$= 0.25 \times 400,000$$

$$= \boxed{100,000}$$

4 Units Needed for Target Profit of \$90,000

Step 1: Write the equation

$$\text{Required units} = \frac{\text{Fixed Cost} + \text{Target Profit}}{\text{CM per unit}}$$

$$= \frac{240,000 + 90,000}{15}$$

$$= \frac{330,000}{15} = \boxed{22,000 \text{ units}}$$

5 Margin of Safety

Step 1: Write formula (dollars)

$$\text{Margin of Safety} = \text{Total sales} - \text{Break-even Sales}$$

$$= 1,200,000 - 960,000$$

$$= \boxed{240,000}$$

Margin of Safety Percentage

Step 1: Formula

$$\text{MOS \%} = \frac{\text{Margin of Safety}}{\text{Actual Sales}}$$

$$= \frac{240,000}{1,200,000}$$

$$= \boxed{20\%}$$

6 Degree of Operating Leverage (DOL)

(a) DOL at Present Sales

Degree of operating leverage measures how sensitive net operating income is to a change in sales.

Step 1: Formula

$$\text{DOL} = \frac{\text{Contribution Margin}}{\text{Net Operating Income}}$$

$$= \frac{300,000}{60,000}$$

$$= \boxed{5}$$

(b) % Increase in NOI if Sales Increase by 8%

Step 1: Equation

$$\begin{aligned}\% \Delta \text{NOI} &= \text{DOL} \times \% \Delta \text{Sales} \\ &= 5 \times 8\% \\ &= \boxed{40\%}\end{aligned}$$

(c) Verification Using New Contribution Income Statement

Step 1: Compute new sales

Factor out original sales:

$$= \text{Original Sales} \times (1 + 0.08)$$

Add 1 and 0.08

$$1 + 0.08 = \boxed{1.08}$$

$$\text{New Sales} = 1,200,000 \times 1.08 = 1,296,000$$

Step 2: Compute new contribution margin

(CM ratio = 25%)

$$\text{Contribution Margin} = 25\% \times 1,296,000 = 324,000$$

Step 3: Compute new NOI

$$\begin{aligned}\text{NOI} &= \text{CM} - \text{Fixed Cost} \\ &= 324,000 - 240,000 = 84,000\end{aligned}$$

Step 4: Verify percentage increase

$$\begin{aligned}\text{Increase in NOI} &= 84,000 - 60,000 = 24,000 \\ \% \text{ Increase} &= \frac{24,000}{60,000} = 40\%\end{aligned}$$

 **Verified**

7 QUESTION 7

7(a) Projected Contribution Format Income Statement

Step 1: Identify changes

- Selling price per unit = \$60
- Variable cost increases by \$3

$$\text{New VC} = 45 + 3 = 48$$

- Contribution per unit:

$$60 - 48 = 12$$

- Fixed costs decrease by \$30,000:

$$240,000 - 30,000 = 210,000$$

- Sales increase by 20%:

$$20,000 \times 1.20 = 24,000 \text{ units}$$

Projected Income Statement

Item	Total	Per Unit	% of Sales
Sales (24,000 units)	\$1,440,000	\$60	100%
Variable expenses	1,152,000	\$48	80%
Contribution margin	288,000	\$12	20%
Fixed expenses	210,000	—	—
Net operating income	78,000	—	—

7(b) New Break-Even Point

Step 1: Write formula (units)

$$\text{BEP (units)} = \frac{\text{Fixed Cost}}{\text{CM per unit}}$$

$$= \frac{210,000}{12}$$

$$= \boxed{17,500 \text{ units}}$$

Break-even in Sales Dollars

Step 1: Formula

$$\begin{aligned} \text{BEP (sales)} &= \text{BEP units} \times \text{Selling price} \\ &= 17,500 \times 60 \\ &= \boxed{1,050,000} \end{aligned}$$

c) The answer is **YES**, and the reason is based on **profit and margin of safety**, not just break-even point.

♦ **Step-by-Step Explanation**

Net Operating Income increases (main reason)

- Present NOI = **\$60,000**
- New NOI = **\$78,000**

$$\text{Increase in NOI} = 78,000 - 60,000 = 18,000$$

Break-even point increases (this is NOT bad by itself)

- Old break-even = **16,000 units**
- New break-even = **17,500 units**

Yes, break-even is **higher**, but **higher break-even does NOT automatically mean more risk**.

Margin of Safety after the change

Formula (write this in exam):

$$\text{Margin of Safety} = \text{Total Sales} - \text{Break-even Sales}$$

Substitute values:

$$= 1,440,000 - 1,050,000$$

Calculate:

$$= \boxed{390,000}$$

Compare with old margin of safety

- Old margin of safety = **\$240,000**
- New margin of safety = **\$390,000**

Margin of safety has increased, which means:

- Less risk
- More cushion before losses occur

■ Definitions of Important CVP Terms

1 Variable Cost

Definition:

Variable cost is a cost that **changes directly with the level of production or sales**.

Example:

If raw material costs **\$45 per unit**, producing more units increases total variable cost.

2 Fixed Cost

Definition:

Fixed cost is a cost that **remains constant in total regardless of output level** within a relevant range.

Example:

Factory rent of **\$240,000 per year** remains the same whether 5,000 or 20,000 units are produced.

3 Selling Price

Definition:

Selling price is the **amount charged to customers for each unit sold**.

Example:

If total sales are \$1,200,000 for 20,000 units, selling price = **\$60 per unit**.

4 Contribution Margin (CM)

Definition:

Contribution margin is the **amount remaining from sales after deducting variable costs**, which contributes toward covering fixed costs and profit.

Formula:

$$\text{Contribution Margin} = \text{Sales} - \text{Variable Costs}$$

Example:

$$\$60 - \$45 = \$15 \text{ per unit}$$

5 Contribution Margin Ratio (CM Ratio)

Definition:

CM ratio is the **percentage of sales revenue that is available to cover fixed costs and profit**.

Formula:

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

Example:

$$\$300,000 \div \$1,200,000 = 25\%$$

6 Variable Expense Ratio

Definition:

Variable expense ratio is the **percentage of sales that represents variable costs**.

Formula:

$$\text{Variable Expense Ratio} = \frac{\text{Variable Costs}}{\text{Sales}}$$

Example:

$$\$900,000 \div \$1,200,000 = 75\%$$

7 Break-Even Point (BEP)

Definition:

Break-even point is the **level of sales where total revenue equals total cost**, resulting in **no profit and no loss**.

Example:

At **16,000 units**, total contribution equals fixed costs.

8 Margin of Safety

Definition:

Margin of safety is the **excess of actual or budgeted sales over break-even sales**.

Formula:

$$\text{MOS} = \text{Actual Sales} - \text{Break-even Sales}$$

Example:

$$\$1,200,000 - \$960,000 = \$240,000$$

9 Net Operating Income (NOI)

Definition:

Net operating income is the **profit earned from normal business operations after deducting all operating costs**.

Example:

$$\text{Contribution margin } \$300,000 - \text{Fixed cost } \$240,000 = \$60,000$$

10 Degree of Operating Leverage (DOL)

Definition:

Degree of operating leverage measures **how sensitive operating income is to changes in sales**.

Formula:

$$\text{DOL} = \frac{\text{Contribution Margin}}{\text{Net Operating Income}}$$

Example:

$$\$300,000 \div \$60,000 = 5$$

1. Contribution Margin (CM)

Contribution Margin (Total)

$$\text{Contribution Margin} = \text{Sales} - \text{Variable Expenses}$$

Contribution Margin (Per Unit)

$$\text{CM per unit} = \text{Selling Price per Unit} - \text{Variable Cost per Unit}$$

2. Net Income (Contribution Format Income Statement)

$$\text{Net Income} = \text{Contribution Margin} - \text{Fixed Expenses}$$

3. Contribution Margin Ratio (CM Ratio)

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

or (per unit form)

$$\text{CM Ratio} = \frac{\text{CM per unit}}{\text{Selling Price per unit}}$$

4. Increase in Contribution Margin (Using CM Ratio)

$$\text{Increase in CM} = \text{Increase in Sales} \times \text{CM Ratio}$$

5. Break-Even Analysis

A. Break-Even Definition

- At break-even:

$$\text{Net Income} = 0$$

B. Break-Even (Equation Method – Units)

$$\text{Sales} = \text{Variable Expenses} + \text{Fixed Expenses} + \text{Profit}$$

At break-even (Profit = 0):

$$\text{Sales} = \text{Variable Expenses} + \text{Fixed Expenses}$$

Expanded form:

$$\text{Selling Price} \times Q = \text{Variable Cost} \times Q + \text{Fixed Costs}$$

C. Break-Even Units Formula (Shortcut)

$$\text{Break-even units} = \frac{\text{Fixed Expenses}}{\text{CM per unit}}$$

D. Break-Even (Equation Method – Sales Dollars)

$$X = (\text{Variable Expense Ratio})X + \text{Fixed Expenses}$$

or

$$X = \text{Break-even Sales in Dollars}$$

6. Target Profit Analysis

A. Target Profit (Units – Equation Method)

$$\text{Selling Price} \times Q = \text{Variable Cost} \times Q + \text{Fixed Costs} + \text{Target Profit}$$

B. Target Profit (Units – CM Approach)

$$\text{Units required} = \frac{\text{Fixed Expenses} + \text{Target Profit}}{\text{CM per unit}}$$

C. Target Profit (Sales Dollars)

$$\text{Required Sales} = \frac{\text{Fixed Expenses} + \text{Target Profit}}{\text{CM Ratio}}$$

7. Margin of Safety (MOS)

A. Margin of Safety (Amount)

$$\text{Margin of Safety} = \text{Total Sales} - \text{Break-even Sales}$$

B. Margin of Safety (Ratio / Percentage)

$$\text{MOS Ratio} = \frac{\text{Margin of Safety}}{\text{Actual Sales}}$$

8. Degree of Operating Leverage (DOL)

$$\text{Degree of Operating Leverage} = \frac{\text{Contribution Margin}}{\text{Net Income}}$$

9. Effect of Sales Change on Net Income (Using DOL)

$$\% \text{ Change in Net Income} = \% \text{ Change in Sales} \times \text{DOL}$$